Title: Food Delivery Process Optimisation and Online Service Enhancement System: Foodhub UK

Dataset: https://www.kaggle.com/datasets/denkuznetz/food-delivery-time-prediction

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D1 – Company Background

1. Organisation Overview

Foodhub UK is an Internet-based food ordering and delivery company that is rapidly expanding in the United Kingdom. Foodhub, unlike its competitors, is a business with zero-commission, where restaurants pay a flat monthly payment as opposed to a percentage of each order. This is a rare approach that has helped the company to win thousands of independent restaurants in cities like Manchester, Birmingham, London, Liverpool, and Leicester (Bhatti and Alawad, 2024).

Foodhub offers an online platform that allows customers to browse menus, order goods, and even deliveries with the use of a mobile and web-based system. The company has the objective of providing affordable, reliable, and user-friendly food ordering services and assisting local restaurants to use cost-effective digital solutions (Keeble et al., 2022). Its business is based on the work of the customers, restaurants, delivery partners and internal dispatching systems.

Figure 1 Foodhub Uk

Source: https://foodhub.co.uk/

2. Sector and Market

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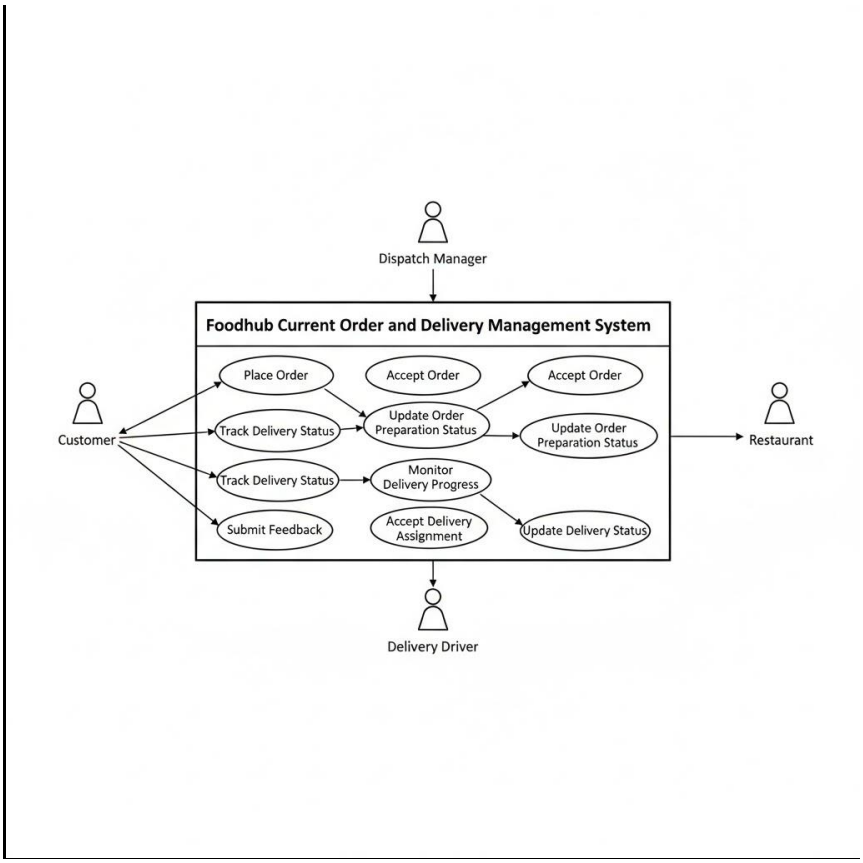

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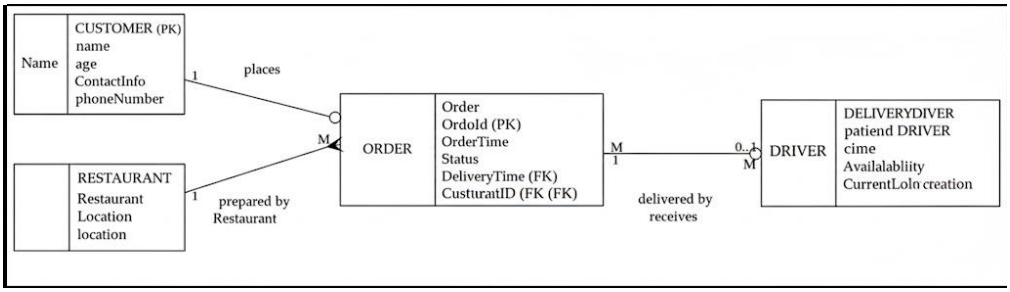

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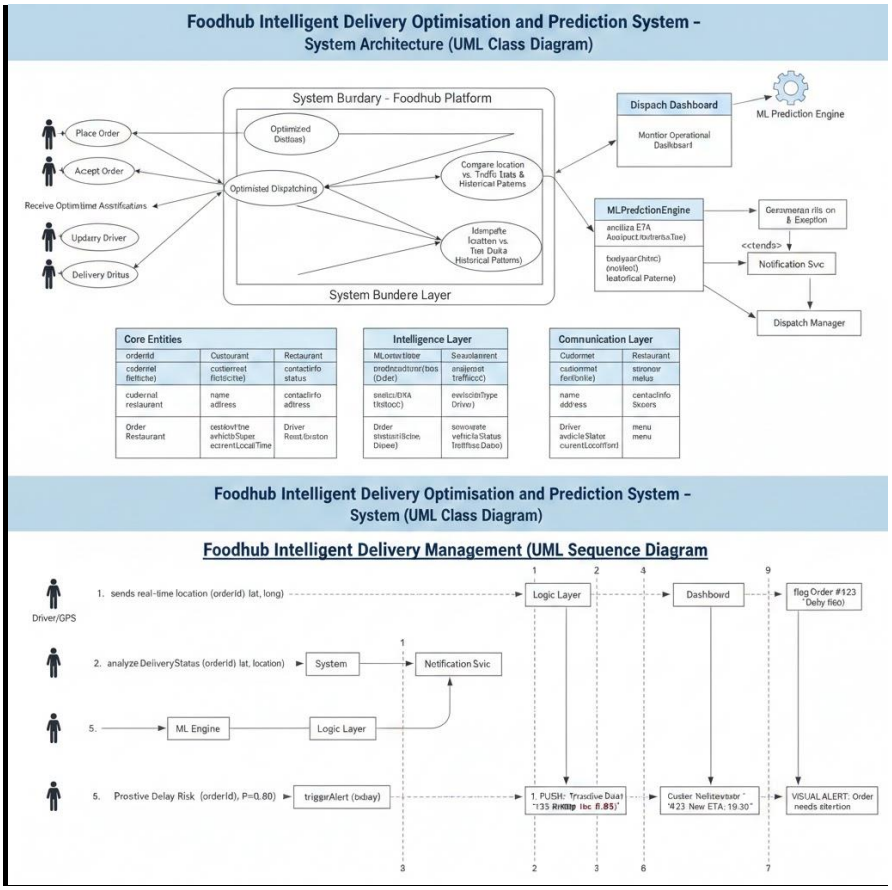
Foodhub Intelligent Delivery Optimisation and Prediction System –
System Architecture (UML Class Diagram)
System Burdary - Foodhub Platform
Place Order
Acept Order
Updarry Driver
Delivery Dritus
Optimized Dstoas)
Optimisted Dispatching
Compare location vs. Tndfis Tsats & Historical Paterns
System Bundere Layer
Dispach Dashboard
Montior Operational Daslkbsarl
ML Predictlon Engine
MLPredctionEngine
Notification Svc
Dispatch Manager
Core Entities
Intelligence Layer
Comnunication Layer
Foodhub Intelligent Delivery Optimisation and Prediction System –
System (UML Class Diagram)
Foodhub Intelligent Delivery Management (UML Sequence Diagram
Driver/GPS
1. sends real-time location (orderid) lat, long)
Logic Layer
Dashbowd
2. analyze DeliveryStatus (orderid) lat, location)
System
Notification Svic
ML Engine
Logic Layer
5. Prostive Delay Risk (orderid), P=0.80)
triggerAlert (bcbay)
VISUAL ALERT: Order needs ntertion

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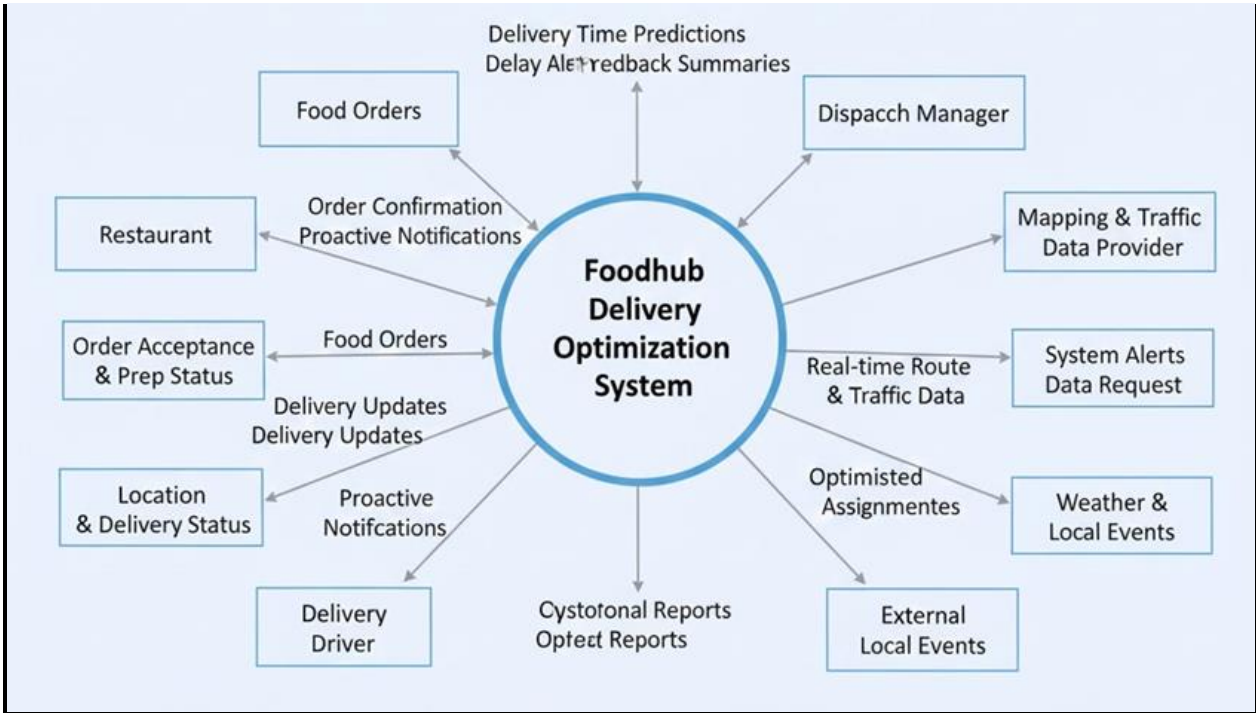

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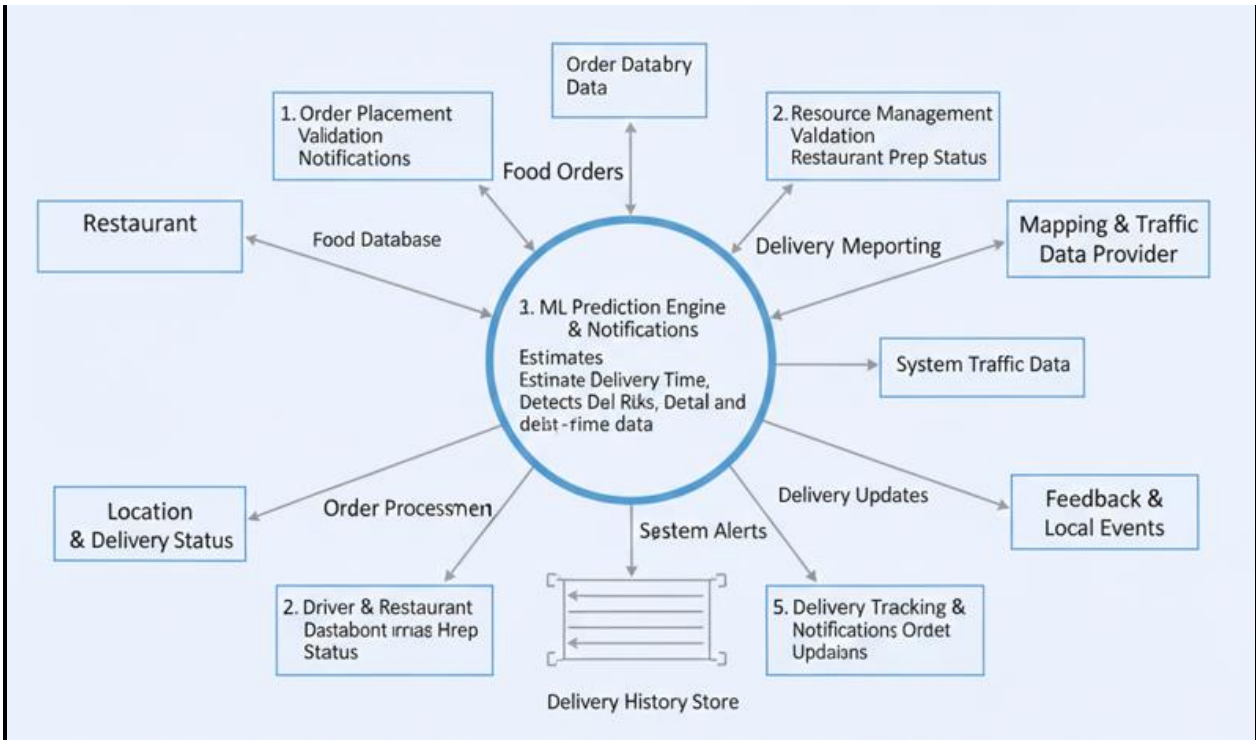
Order Databry Data
1. Order Placement Validation Notifications
Food Orders
2. Resource Management Valdation Restaurant Prep Status
Restaurant
Food Database
Delivery Meporting
Mapping & Traffic Data Provider
1. ML Prediction Engine & Notifications
Estimates
Estinate Delivery Time, Detects Del Riks, Detal and deist-fime data
System Traffic Data
Location & Delivery Status
Order Processnen
Delivery Updates
Feedback & Local Events
Sestem Alerts
2. Driver & Restaurant Dastabont irnas Hrep Status
5. Delivery Tracking & Notifications Ordet Updaions
Delivery History Store

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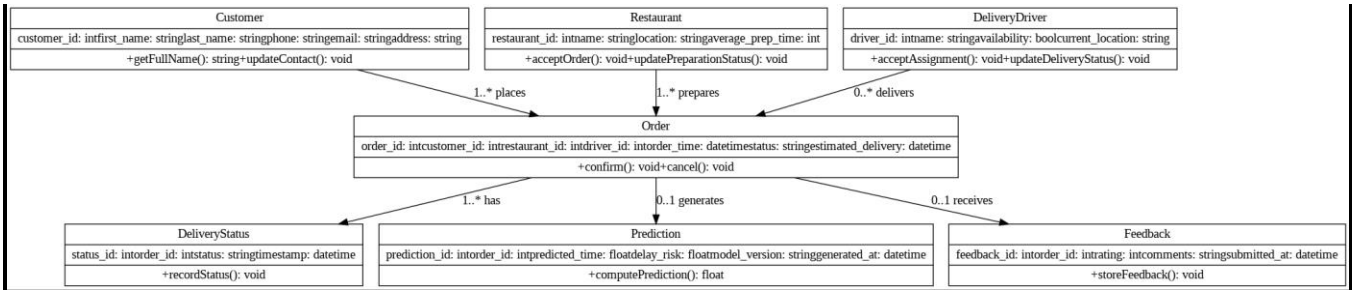

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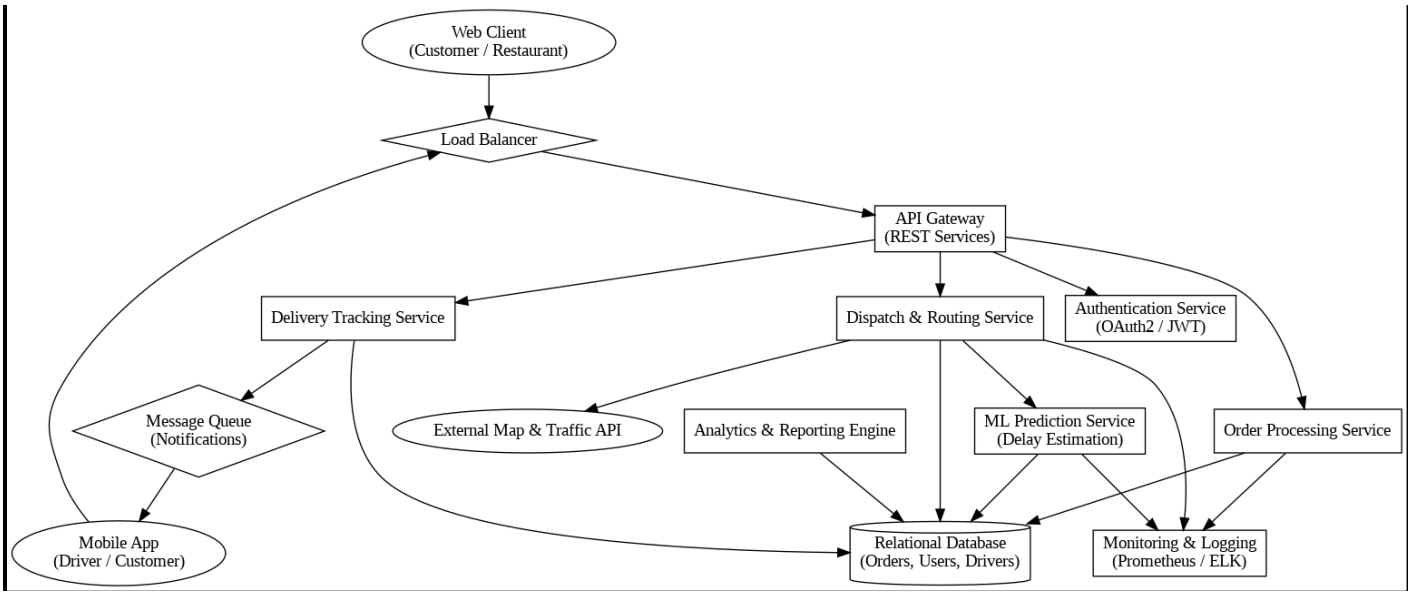

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```
Columns:
Index(['Order_ID', 'Distance_km', 'Weather', 'Traffic_Level', 'Time_of_Day',
       'Vehicle_Type', 'Preparation_Time_min', 'Courier_Experience_yrs',
       'Delivery_Time_min'],
      dtype='object')
```

	Order_ID	Distance_km	Weather	Traffic_Level	Time_of_Day	Vehicle_Type	Preparation_Time_min	Courier_Experience_yrs	Delivery_Time_min
0	522	7.93	Windy	Low	Afternoon	Scooter	12	1.0	43
1	738	16.42	Clear	Medium	Evening	Bike	20	2.0	84
2	741	9.52	Foggy	Low	Night	Scooter	28	1.0	59
3	661	7.44	Rainy	Medium	Afternoon	Scooter	5	1.0	37
4	412	19.03	Clear	Low	Morning	Bike	16	5.0	68

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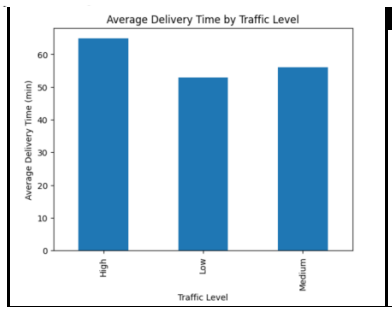
```
Missing values after cleaning:
Order_ID                 0
Distance_km              0
Weather                  0
Traffic_Level            0
Time_of_Day              0
Vehicle_Type             0
Preparation_Time_min     0
Courier_Experience_yrs   0
Delivery_Time_min        0
dtype: int64
```



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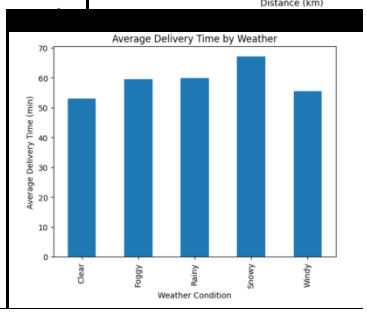

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Random Forest Performance:
MAE  : 7.1429
RMSE: 10.318234635827972
R²   : 0.7624730952437657
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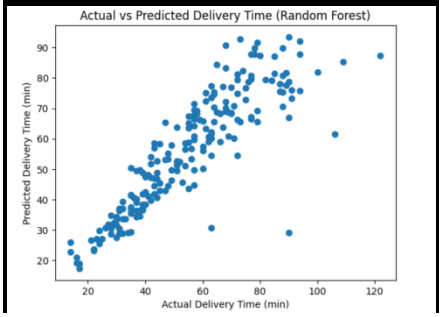

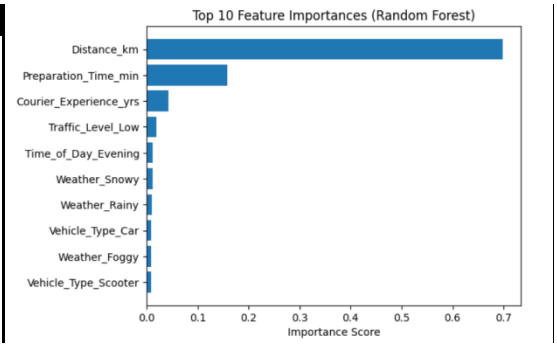

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